See Hidden Insight From Transposition: Multiaxis Feature Aggregation for Aerial Object Detection

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***Abstract*—Recent researches perform aerial object detection from particular perspectives, e.g., small resolution and arbitrary orientations of instances. However, objects with large aspect ratios are actually very common in aerial scenarios, and recent methods seldom carry out their work on the aforementioned topic. In this article, we propose a large aspect ratio object-related method, named see hidden insight from transposition (SHIFT), to elevate the performance of aerial object detection. First, we regard the common convolutional operation as the forward-view feature modeling process, which generates the initially comprehensive descriptors for most aerial objects. This step serves as the foundation for capturing the basic features of different objects. By transposing image features into top- and left-view, our SHIFT can better capture the overall shape and spatial structure of large aspect ratio objects, which may be partially missed or inaccurately represented in the forward-view. Obviously, the top- and left-view features are all strip-like maps, which implicitly express the height and width of objects in the image, rather than the texture and appearance. In order to bridge the representational gaps among the aforementioned multiview features, we further construct an adaptive feature fusion (AFF) module to fuse them in an intelligent way, which models the implicit relationships among these descriptors through the soft attention weights. Our AFF module can be treated as a plug-and-play component, which is end-to-end trainable in the whole detection network. Experimental results on DOTA-v1.0, DOTA-v1.5, and DIOR-R datasets demonstrate that our method achieves superior performance than many recent object detection approaches by a significant margin, including various large aspect ratio related methods.**

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***Index Terms*—Aerial object detection, feature fusion, hidden insight, multiaxis, remote sensing images (RSIs).**

I. Introduction

DIFFERENT from object detection in natural images [1], [2], [3], aerial object detection is much more challenging due to the extremely small resolution and arbitrary orientations of instances. Correspondingly, recent efforts [4], [5], [6], [7], [8], [9], [10], [11] are devoted to addressing the aforementioned limitations. Considering that horizontal candidate proposals are difficult to perceive instances with different angles, methods [12], [13] focus on generating oriented proposals to better perceive the oriented instances by the user-defined or learnable parameters. Moreover, most recent approaches [7], [14], [15], [16] are devoted to modeling rotation-invariant features. By exploring the intrinsic consistency constraints of features before and after rotation, their detectors are enabled to adapt to targets with various rotational orientations. Some recently proposed methods [17], [18] excavate better features to boost the performance of aerial small object detection. They use attention mechanisms and multiscale learning strategies [19], [20] to allocate the computational resources toward important informative parts of the image, which pose great potential for highlighting the key spatial responses and suppressing the redundant ones in the image.

However, as shown in Fig. 1, large aspect ratio instances are actually very common in aerial scenarios. Specifically, objects with a large aspect ratio account for about half of the total number. The potential solution to this topic is seldom performed in existing methods. They only pin their hopes on performing a downsampling operation and sliding the convolutional windows for perceiving narrow instances. Unfortunately, in real scenarios, the aforementioned assumption still hardly meets the detection requirements for aerial targets with large aspect ratios. In this article, we propose a large aspect ratio object-related method, named see hidden insight from transposition (SHIFT), to elevate the performance of aerial object detection. Initially, we regard the common convolution operation as the forward-view feature modeling process, which provides the comprehensive and professional indicators for understanding potential information, such as the scales and distributions of objects. This process performs convolution operations on the plane composed of the height axis (H-axis) and width axis (W-axis). Moreover, our method expands the convolutional planes by modeling image features

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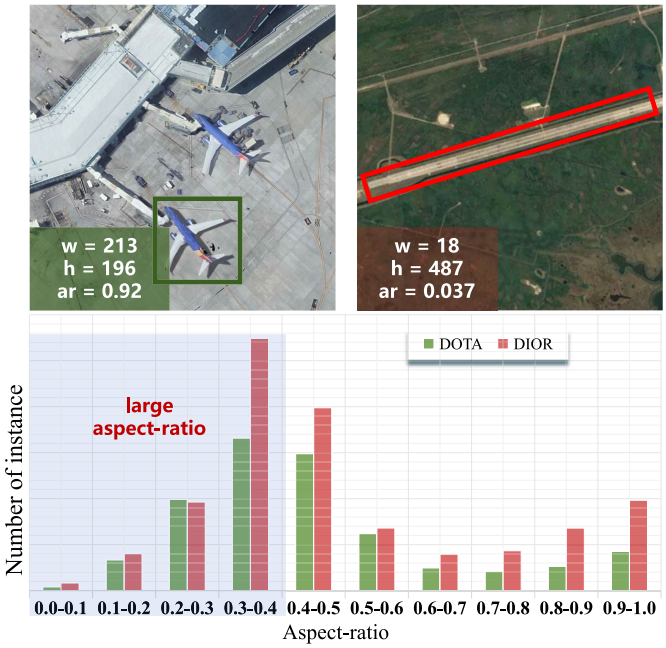


Fig. 1. Aerial objects with different aspect ratios. Specifically, aspect ratio (ar) is obtained by dividing the short side by the long side of the object. Therefore, smaller ar denotes a narrower instance. In addition, the number of instances with different aspect ratios in the DOTA and DIOR-R datasets is also visualized.

on the planes composed of the W-axis and channel axis (C-axis) and H-axis and C-axis, which observes target instances from the left and top lenses. The aforementioned multiaxis features not only enrich the feature information but also enable the detection network to build more comprehensive understandings of objects, especially those with large aspect ratios. Finally, we fuse the aforementioned multiaxis features via the constructed adaptive feature fusion (AFF) module. Our AFF can be integrated into the ResNet seamlessly and is end-to-end trainable along with the whole detection network.

In summary, the main contributions of this article are as follows.

1) We promote aerial object detection task by a particular topic: large aspect ratio, and propose a multiaxis feature aggregation module, named SHIFT, to explicitly achieve more comprehensive understandings of different instances in aerial scenarios.
2) We construct an AFF module to model interactions among the multiaxis descriptors through the soft attention weights, which not only bridge the gaps, but also fuse them in an intelligent way.
3) Extensive experiments demonstrate that our method achieves superior performance to many state-of-the-art approaches, especially for instances with large aspect ratios. Importantly, we for the first time evaluate the performance of aerial object detection under the aspect ratio specified AP_{ar} criterion (Section IV).

II. Related Works

A. Aerial Object Detection

Aerial object detection is much more challenging than that in natural scenes, due to the extremely small resolution, arbitrary orientations, and large aspect ratios of the target instances. Most recent researches elevate the performance of the aerial object detection task from two perspectives: small object detection and rotation-invariant feature learning.

Oriented R-CNN [12] was dedicated to generating high-quality oriented anchors to better cover the target instances. RoI transformer [13] utilized multiple fully-connected layers to transfer horizontal candidate boxes into the oriented ones, and then the distorted instance-level features were fed into the category classification and box regression networks. Object-wise semantic representation (OWSR) [16] and object-wise rotation-invariant semantic representation (ORSR) [5] focused on excavating the potential constraints inside the detection network to construct auxiliary tasks, and modeled the intrinsic consistency information of image features before and after rotation. DRN [21] proposed a dynamic refinement network, which constructed a feature selection module (FSM) and a dynamic refinement head (DRH) to make adjustments to receptive fields according to the forms of target objects. The constructed DRH module empowered the detection network to dynamically refine the prediction in an instance-aware manner. In addition, AO2-DETR [22] modified the base DETR [23] to perform the object detection task in remote sensing. Through integrating the transformer framework with convolutional neural networks (CNNs), AO2-DETR boosted detection performance from the perspective of feature diversity, which can be regarded as a variant of feature enhancement learning.

However, the convolutional operations involved in the aforementioned approaches only model the instance descriptors from the forward-view. The detection network pins its hope on performing multiple downsampling layers to observe the complete object areas, which is still ineffective for instances with large aspect ratios. Instead, our approaches model the image indicators from different geometric views to build more comprehensive understandings of objects, which elevates the detection performance for large aspect ratio instances.

B. Solutions for Large Aspect Ratio Object

Although objects with large aspect ratios are very common in aerial scenarios, there are limited works to perform aerial object detection on this topic. Most existing large aspect ratio object detection methods can be divided into five categories: designing irregular kernel [4], constructing large kernel [6], modeling long-range correlation via self-attention (SE) [15], [24], [25], [26] or dynamic sampling [21], [27], [28], and introducing intersection over union (IoU) and angle constraints [29], [30], [31], [32], [33], [34]. Specifically, SE [24] with its variants [15], [25], [26], and dynamic sampling [21], [27], [28] all focused on extending the receptive fields to large aspect ratio targets. However, due to the lack of explicit supervisory guidance, the aforementioned methods struggle to precisely control the direction and extent of perception range expansion and tend to suffer from insufficient correlation modeling when dealing with extremely slender targets. Therefore, the most recent methods introduced the IoU and angle constraints in OOD. For example, the task interleaving and orientation estimation detector (TIOE-Det) [30] presented a progressive orientation estimation (POE) strategy, which approximated

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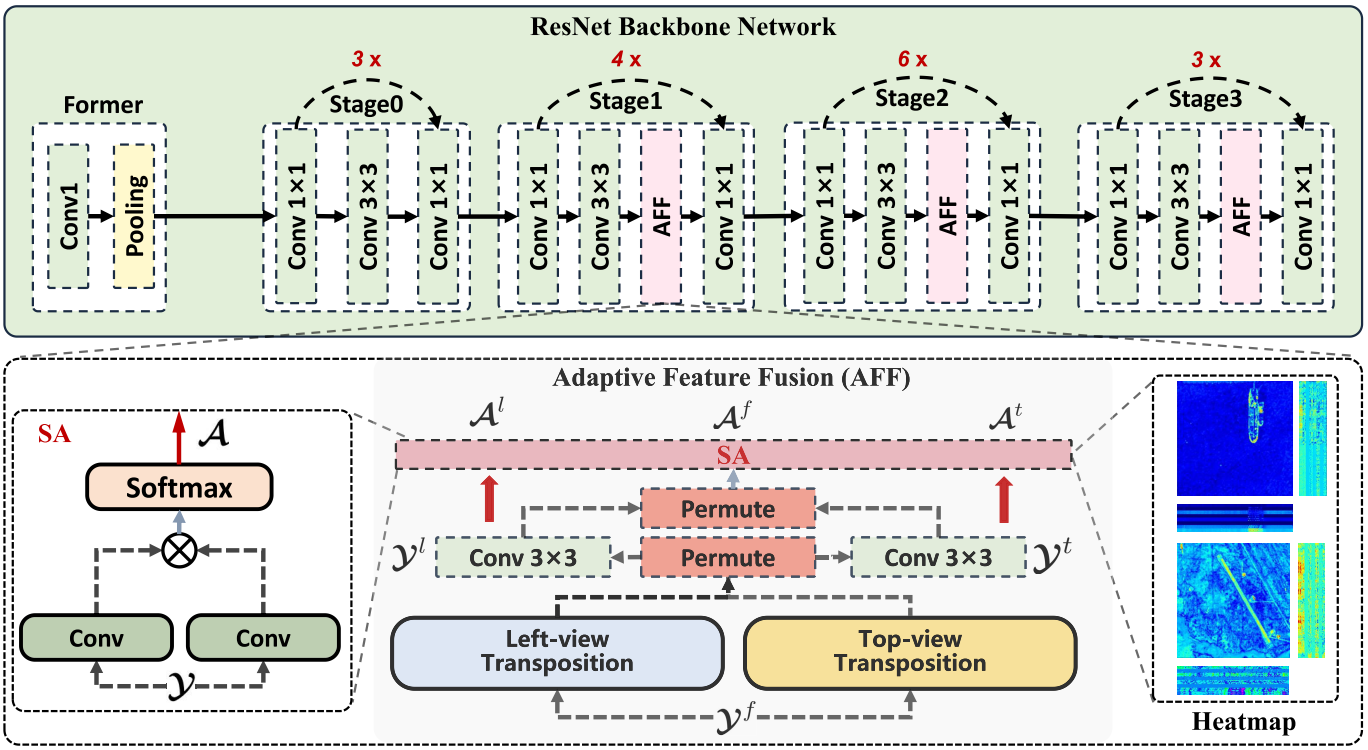


Fig. 2. Overall of our SHIFT. The AFF module is integrated with different ResBlocks of the ResNet backbone network. After attaining the top-view and left-view features $\mathcal{Y}^l$, $\mathcal{Y}^t$ by transposing the forward-view feature $\mathcal{Y}^f$, our AFF aims to bridge the representational gaps among the aforementioned multiaxis features and fuse them in an intelligent way.

the orientations of objects using n-ary codes. Notably, the POE strategy greatly enhanced the detection capability for objects with large aspect ratios by virtue of precise orientation prediction. Starting from the perspective of data augmentation, noisy bounding box (NBBOX) [31] simulated the scaling, translation, and rotation of images to generate diverse label variations, which encouraged the training of models that are robust to changes in bounding box aspect ratios. The aspect-ratio-guided (ARG) [32] improved the label assignment criteria (IoU threshold) in the detection network, and the ARG IoU loss was constructed to adjust angle regression weighting adaptively.

However, the aforementioned methods are still limited to performing convolution operations on the plane composed of the H-axis and W-axis. They struggle to precisely control the direction and extent of perception range expansion due to the lack of supervisory information. Moreover, these methods exhibit an imbalance in perceiving large aspect ratio objects, as excessive attentions focus on angle prediction. In contrast, our method expands the convolutional planes by modeling image features on the planes composed of the W-axis and C-axis and H-axis and C-axis. Therefore, our SHIFT not only eliminates the reliance on supervisory information for sampling points but also focuses all computing resources on modeling the length and width information of slender targets.

III. Proposed Method

As illustrated in Fig. 2, we give the overall of our SHIFT framework from three aspects: 1) modeling hidden insights from diverse lenses of objects in Section III-A; 2) analyzing the hidden insights in Section III-B; and 3) fusing the multiaxis features in an intelligent manner in Section III-C.

A. See Hidden Insight From Transposition

We first define the convolutional operation performed on the plane composed of the H-axis and W-axis as the forward-view feature modeling process. This operation provides a comprehensive and professional indicator for understanding most aerial instances. Given the input feature as $\mathcal{X} \in \mathcal{R}^{\mathcal{C}\times\mathcal{W}\times\mathcal{H}}$, the aforementioned operation can be summarized by

$$\mathcal{Y}^f = W^f \cdot \mathcal{X} \tag{1}$$

where $\mathcal{Y}^f \in \mathcal{R}^{\mathcal{C}\times\mathcal{W}\times\mathcal{H}}$. W^f denotes the forward-view convolutional layer, which performs dimension-preserving feature transformation. Obviously, the receptive field of W^f cannot cover the global view of large aspect ratio objects. Irregular convolution and deformable convolutional network (DCN) [27] alleviate the perceptual limitations for elongated objects by adopting fixed or parameterized adjustments to sampling points, yet the receptive window of DCN remains constrained by the size of the original convolution kernel (e.g., $3 \times 3 = 9$), thus failing to perceive the entire target region in a single step.

Our motivation emerged as the forward-view convolution operation only pins its hopes on multiple downsampling layers to perceive the complete instance area, but it is still ineffective for large aspect ratio objects. Therefore, our method observes objects from other lenses: expanding the original convolutional plane (H-axis and W-axis) and

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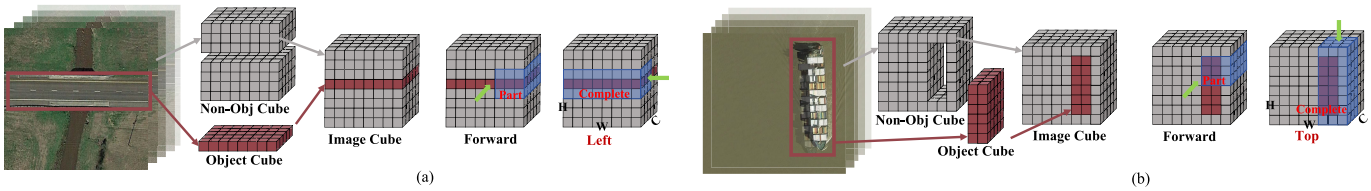


Fig. 3. Effectiveness of our proposed left and top-view convolutions. The green arrow denotes the convolutional direction, and the blue region represents the receptive field. Given the input features with multiple dimensions, the left-view and top-view modeling pose great potential to perceive horizontal and vertical elongated "Bridge" and "Ship" instances. (a) Left view for vertical instance. (b) Top view for horizontal instance.

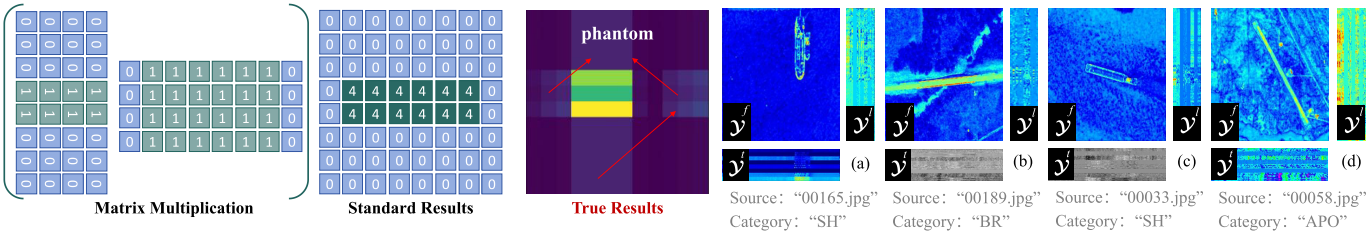


Fig. 4. Analysis of the introduced hidden insights. The comparisons between the standard map (calculated by matrix multiplication) and the true predicted results are illustrated. (a)–(d) $\mathcal{Y}^f$, $\mathcal{Y}^l$, and $\mathcal{Y}^t$ of different images in DIOR-R dataset.

modeling image features on the planes composed of the W-axis and C-axis and H-axis and C-axis.

Based on the aforementioned analysis, we first transfer $\mathcal{X}$ by different $permute\mathcal{P}(\cdot)$ operations as

$$\mathcal{X}^w = \mathcal{P}_{1,0,2}(\mathcal{X}), \mathcal{X}^h = \mathcal{P}_{2,0,1}(\mathcal{X}) \tag{2}$$

where $_{1,0,2}$ and $_{2,0,1}$ denote two dimension permutation operations. $\mathcal{X}^w \in \mathcal{R}^{\mathcal{W}\times\mathcal{C}\times\mathcal{H}}$ and $\mathcal{X}^h \in \mathcal{R}^{\mathcal{H}\times\mathcal{C}\times\mathcal{W}}$ denote the transferred features. Therefore, the multiaxis feature extraction can be simply modified (1) as

$$\mathcal{Y}^l = W^l \cdot \mathcal{X}^w, \mathcal{Y}^t = W^t \cdot \mathcal{X}^h \tag{3}$$

where W^l and W^t denote left-view and top-view convolutional layers, which also perform dimension-preserving feature transformations. $\mathcal{Y}^l \in \mathcal{R}^{\mathcal{W}\times\mathcal{C}\times\mathcal{H}}$ and $\mathcal{Y}^t \in \mathcal{R}^{\mathcal{H}\times\mathcal{C}\times\mathcal{W}}$ denote the corresponding multiaxis features. Before feeding these features into the following AFF module, we also use a pair of $permute\mathcal{P}(\cdot)$ operations to align their dimensions with the original $\mathcal{Y}^f$.

As illustrated in Fig. 3, we use the red rectangles to locate the bridge and ship instances in images. The extracted image features contain dozens or even hundreds of channels, and each channel presents a grayscale image. Notably, we visualize a feature as a Rubik's Cube-like 3-D cube. The background and target regions can be stripped from the complete cube and represented by the gray and red 3-D subcubes. Herein, the complete image cube is formed by assembling these two types of subcubes in an inserted manner. It can be clearly seen that both horizontally slender "Bridge (BR)" and vertically elongated "Ship (SH)" are better perceived by the proposed top-view and left-view convolutions.

B. Analysis of Hidden Insights

In this section, we give a detailed analysis of the hidden insights extracted by our SHIFT. First, forward-view image feature $\mathcal{Y}^f$ can be separated into the object region $\mathcal{O}$ and nonobject region $\mathcal{NO}$ as

$$\mathcal{Y}^f(\mathcal{C},\mathcal{W},\mathcal{H}) = \mathcal{O}(\mathcal{C},w_o,h_o) \begin{matrix}\leftrightarrow\\ \updownarrow\end{matrix} \mathcal{NO}(\mathcal{C},w_{no},h_{no}). \tag{4}$$

As illustrated in Fig. 3, we use different Rubik's Cube-like 3-D cubes to represent the nonobject and object features, respectively, and these cubes are stitched together $\leftrightarrow$ / $\updownarrow$ to construct the complete image features. $\mathcal{O}$ has the same channel as $\mathcal{Y}^f$, and w_o, h_o are smaller than $\mathcal{W}, \mathcal{H}$. Specifically, our SHIFT observes image features from diverse lenses. Obviously, the top- and left-view features are all strip-like maps, which implicitly express the height and width of objects in images. We first use two one-zero metrics, for example, in Fig. 4 to illustrate the aforementioned strip-like maps. It can be clearly seen that we can simply reconstruct the positional indicator of original objects via matrix multiplication. However, as illustrated in Fig. 4, the result of matrix multiplication is more like a positional indicator, not an instance descriptor, but further raises the problem of phantom.

Considering that features in multiple channels represent different cues in the image, the problem of phantoms will be aggravated by the matrix multiplication. As illustrated in Fig. 4(a)–(d), we further visualize the $\mathcal{Y}^f$, $\mathcal{Y}^l$, and $\mathcal{Y}^t$ of different images in DIOR-R dataset. Notably, top and left-view features implicitly encode the height and width of objects in images, as opposed to textures and appearances. These textures and appearances, however, are precisely what forward-view features struggle to capture intuitively. Therefore, it is necessary to fuse these multiaxis features in an intelligent way, and this is exactly what our AFF module does.

C. AFF Module

Our AFF module has three inputs $\mathcal{Y}^f$, $\mathcal{Y}^l$, and $\mathcal{Y}^t$. The former one generates the initially comprehensive descriptors for most aerial objects, and the latter two are utilized to

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supplement more generalizable indicators for the forward-view features. The mainstream feature aggregation methods can be divided into three aspects: *add* (ResNet [35]), *concat* (InceptionNet [36]), and adaptive weighting-based methods. Since $\mathcal{Y}^l$ and $\mathcal{Y}^t$ primarily encode the width and height of objects, which inevitably obscure instance textures and details. Therefore, our AFF module follows an attention-based manner to perform the multiaxis feature aggregation. In a standard convolutional layer, the kernel traverses the spatial dimension of the input feature, which can be regarded as a single-view spatial scan. Our proposed SHIFT module extends this scanning process by applying two different permutation operations, and then the convolutional kernel effectively traverses along distinct dimensional arrangements. Therefore, in addition to the original spatial scan, we introduce two extra scanning perspectives, which constitute a multiview feature scanning mechanism.

Specifically, the proposed AFF module calculates the generalized cosine similarity among $\mathcal{Y}^f$ as

$$\mathcal{A}^f = softmax\left(\phi_1\left(\mathcal{Y}^f\right)^T \cdot \theta_1\left(\mathcal{Y}^f\right)\right) \tag{5}$$

where $\phi_1(\cdot)$ and $\theta_1(\cdot)$ denote two different combinations of batch normalization, ReLU, and convolution layers. $\mathcal{A}^f$ denotes the SE [24] weighting matrix. Higher similarity indicates the relationship between two regions in $\mathcal{Y}^f$ is valuable.

Then, we use two different $permute\mathcal{P}(\cdot)$ operations to unify the dimensions of $\mathcal{Y}^f$, $\mathcal{Y}^l$, and $\mathcal{Y}^t$ as

$$\mathcal{Y}^l = \mathcal{P}_{1,0,2}\left(\mathcal{Y}^l\right), \mathcal{Y}^t = \mathcal{P}_{1,2,0}\left(\mathcal{Y}^t\right) \tag{6}$$

where $\mathcal{Y}^f, \mathcal{Y}^l$, and $\mathcal{Y}^t \in \mathcal{R}^{\mathcal{C}\times\mathcal{W}\times\mathcal{H}}$. Therefore, the generalized cosine similarity among $\mathcal{Y}^l$ and $\mathcal{Y}^t$ can be modified (5) as $\mathcal{A}^l = softmax(\phi_2(\mathcal{Y}^l)^T \cdot \theta_2(\mathcal{Y}^l))$ and $\mathcal{A}^t = softmax(\phi_3(\mathcal{Y}^t)^T \cdot \theta_3(\mathcal{Y}^t))$. After defining the similarity among these multiaxis features, $\mathcal{A}^l$ and $\mathcal{A}^t$ serve as the guidance for $\mathcal{A}^f$ to aggregate the hidden insights from other lenses. We merge these SE similarity weights as

$$\mathcal{A} = \mathcal{A}^f + \mathcal{A}^l + \mathcal{A}^t \tag{7}$$

and the final enhanced forward-view features $\hat{\mathcal{Y}}^f$ is achieved by

$$\hat{\mathcal{Y}}^f = \mathcal{A} \cdot \beta\left(\mathcal{Y}^f\right) \tag{8}$$

where $\beta(\cdot)$ represents a combination of batch normalization, ReLU, and convolution layers, which serves as the $K(\cdot)$ function in the SE algorithm.

Finally, we integrate our proposed AFF module with the ResNet [35] backbone. Considering that each ResBlock in ResNet has three convolutional layers, and most recent research [37], [38] are dedicated to modifying the second or third convolutions in ResBlock. Therefore, we also construct the corresponding ablation studies on the aforementioned arrangements in Section IV.

IV. Experiments

A. Datasets

DOTA-v1.0 [39] includes 2806 aerial images with spatial resolution ranging from 800 × 800 to 4000 × 4000. The training, validation, and testing sets contain 1411, 458, and 937 images, respectively. 15 categories of 188 282 instances are annotated in DOTA-v1.0, including plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). Followed by the standard settings in methods [40], [41], both training and validation sets are used for training our models. The verification results on the testing set need to be uploaded to the official server for evaluation.

DOTA-v1.5 [39] has the same number of images as DOTA-v1.0. A new category, named container crane (CC), is annotated in DOTA-v1.5. Meanwhile, a total of 402 089 instances are labeled. We also use training and validation sets to train our models. The verification results on the testing set are uploaded to the official server for evaluation.

DIOR-R [42] contains 23 463 images with 192 518 annotated instances. A total of 20 categories: airplane (APL), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), airport (APO), dam (DAM), expressway service area (ESA), expressway toll station (ETS), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), windmill (WM), golf field (GF), and ground track field (GTF) are labeled in DIOR-R. Followed by the standard settings in methods [43], [44], we use 11 725 samples for training and the rest 11 738 images are used for evaluation.

B. Implementation Details

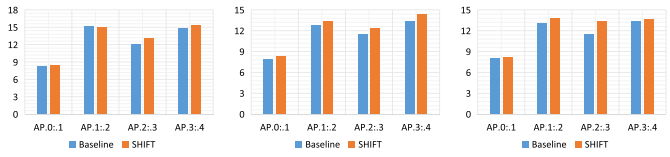
Followed by the recent approaches [4], [6], [45], we build our SHIFT based on the ReDet [7] framework after the following careful considerations. First, when the object is in a horizontal or vertical state, our SHIFT can better capture its width and height information. However, instances in remote sensing images (RSIs) have arbitrary orientations. ReDet adopts the general E(2)-equivariant steerable CNNs (E2CNN), wherein a single convolution kernel outputs features with a specified number of channels, and the number of channels is equal to the preset number of orientations. Therefore, "kernel rotates while the image stays still," involved in ReDet, poses great potential to alleviate the aforementioned limitations and support our SHIFT.

For DOTA, images are cropped into 1024 × 1024 patches with an overlap of 200 between two adjacent patches. The SGD optimizer with momentum 0.9, weight decay 0.0001, and learning rate 0.005 is introduced in DOTA and DIOR-R datasets to optimize our models.

C. Ablation Studies

1) AP Under Different Aspect Ratios (0.0:1.0): In this section, we divide the aspect ratios of the predicted bounding boxes into ten intervals at a step of 0.1, and evaluate the AP within each aspect ratio interval range, respectively. As listed in Table I, $AP_{.0:.1}$ denotes that the AP is evaluated by the instances that aspect ratios range from 0.0 to 0.1. Predictions where the IoU between the predicted boxes and GT boxes is

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(a) (b) (c)

Fig. 5. Improvements with different IoU threshold. (a)–(c) *ovthresh* are set to 0.4, 0.5, and 0.6, respectively.

TABLE I

AP$_{ar}$ CRITERION UNDER DIFFERENT RANGES OF ASPECT RATIOS ON DIOR-R. WE DIVIDE THE ASPECT RATIOS OF THE PREDICTED BOUNDING BOXES INTO TEN INTERVALS AT A STEP OF 0.1, AND EVALUATE THE AP WITHIN EACH ASPECT RATIO INTERVAL RANGES (SINGLE RANGE). IN ADDITION, WE ALSO EVALUATE AP$_{ar}$ UNDER ONE CERTAIN RANGE OF ASPECT RATIO (MULTIRANGES). FOR EXAMPLE, AP$_{\leq 0.2}$ DENOTES THE AP IS CALCULATED BY THE PREDICTED BOXES THAT ASPECT RATIOS ARE LOWER THAN 0.2. TOTAL COLUMN DENOTES THE SUMMATION OF IMPROVEMENTS

Single Range						
$ovthresh = 0.4$						
Method	AP$_{.0:.1}$	AP$_{.1:.2}$	AP$_{.2:.3}$	AP$_{.3:.4}$	AP$_{.4:.5}$	Gains
Baseline	8.3	15.2	12.0	14.8	16.5	–
SHIFT	**8.5** $\uparrow_{0.2}$	**14.9** $\downarrow_{0.3}$	**13.0** $\uparrow_{1.0}$	**15.3** $\uparrow_{0.5}$	**16.6** $\uparrow_{0.1}$	**1.5**
	AP$_{.5:.6}$	AP$_{.6:.7}$	AP$_{.7:.8}$	AP$_{.8:.9}$	AP$_{.9:1.0}$	Total
Baseline	13.8	12.3	11.7	12.7	15.1	–
SHIFT	14.2 $\uparrow_{0.4}$	12.0 $\downarrow_{0.3}$	12.0 $\uparrow_{0.3}$	13.0 $\uparrow_{0.3}$	15.7 $\uparrow_{0.6}$	1.3
$ovthresh = 0.5$						
Method	AP$_{.0:.1}$	AP$_{.1:.2}$	AP$_{.2:.3}$	AP$_{.3:.4}$	AP$_{.4:.5}$	Total
Baseline	7.9	12.8	11.4	13.3	16.0	–
SHIFT	**8.3** $\uparrow_{0.4}$	**13.4** $\uparrow_{0.6}$	**12.3** $\uparrow_{0.9}$	**14.3** $\uparrow_{1.0}$	**16.1** $\uparrow_{0.1}$	**2.1**
	AP$_{.5:.6}$	AP$_{.6:.7}$	AP$_{.7:.8}$	AP$_{.8:.9}$	AP$_{.9:1.0}$	Total
Baseline	13.7	12.2	11.6	12.6	14.7	–
SHIFT	14.2 $\uparrow_{0.4}$	11.9 $\downarrow_{0.3}$	11.6 $\uparrow_{0.0}$	12.4 $\downarrow_{0.2}$	15.1 $\uparrow_{0.4}$	0.4
Multi Ranges						
Method	AP$_{\leq 0.1}$	AP$_{\leq 0.2}$	AP$_{\leq 0.3}$	AP$_{\leq 0.4}$	AP$_{\leq 0.5}$	Total
Baseline	7.9	14.0	16.4	24.5	34.2	–
SHIFT	**8.3** $\uparrow_{0.4}$	**14.5** $\uparrow_{0.5}$	**18.0** $\uparrow_{1.6}$	**25.1** $\uparrow_{0.6}$	**34.7** $\uparrow_{0.5}$	3.6
	AP$_{\leq 0.6}$	AP$_{\leq 0.7}$	AP$_{\leq 0.8}$	AP$_{\leq 0.9}$	AP$_{\leq 1.0}$	Total
Baseline	40.5	45.8	51.1	56.6	66.0	–
SHIFT	40.9 $\uparrow_{0.4}$	46.9 $\uparrow_{1.1}$	51.5 $\uparrow_{0.4}$	57.9 $\uparrow_{1.3}$	67.9 $\uparrow_{1.9}$	5.1

greater than *ovthresh* are denoted as positive predictions. It can be clearly seen that the proposed SHIFT significantly boosts the detection performance of large aspect ratio objects, which achieves 0.4%, 0.6%, 0.9%, and 1.0% gains over the baseline under 0.5 *ovthresh*, respectively. Importantly, compared with the instances with regular aspect ratios, our method greatly improves the detection performance of large aspect ratio related objects, which achieves 1.5% (versus 1.3%) and 2.1% (versus 0.4%) AP gains. As illustrated in Fig. 5, compared with the baseline method, our proposed SHIFT always achieves significant mAP improvements under different IoU thresholds.

In addition, we further evaluate AP$_{ar}$ under a certain range of aspect ratio (multiranges). Our SHIFT is always superior to baseline by significant margins, which achieves the best 1.6% AP gains in terms of AP ≤ 0.3 for large aspect ratio objects. The aforementioned comparisons prove that our SHIFT has a superior locating and identification ability on large aspect ratio objects. As shown in Fig. 6, for large aspect ratio objects, the spatial responses in the original feature maps tend to be spatially dispersed or blurred at boundaries, exhibiting weak feature responses and indistinct structures. In contrast, the feature maps enhanced by the SHIFT module demonstrate more concentrated and coherent spatial activation, with significantly enhanced representational capacity for both the main body and boundaries of the objects. This further validates the effectiveness of the proposed method in modeling large aspect ratio instances.

2) Compared to Other Aspect Ratio Related Methods: We compare our SHIFT with many aspect ratio-related approaches in Table II. The compared methods can be divided into five categories, including dynamic sampling (DRN [21], DCN [27], and DAL [28]), irregular kernel (ARC [4]), large kernel (LSKNet-S [6]), IoU and angle loss (R^{3}Det [29], KLD [34], and B2V [33]), and SE (ARS-DETR [25], SE [24], and FDOL [15]).

Specifically, SE with its variants and dynamic sampling, all focused on extending the receptive fields to large aspect ratio targets, and the best competitor, FDOL [15], achieves 77.22% mAP. LSKNet-S [6] retrains its backbone network on the ImageNet-1K dataset for 300 epochs, and adjusts the shape of convolution kernels to optimize target perception, which ultimately achieves a mAP of 77.49%. Moreover, to compare against our proposed SHIFT, we individually integrate the DCN [27], SE [24] with its variants [15], irregular kernel with size of 1×5, and large kernel with size of 5×5 into the same baseline method ReDet [7]. As listed in Tables II and III, large and irregular convolutional kernels pose great potential to perceive large aspect ratios, which achieve 0.22% and 0.67% mAP gains over the baseline, respectively. FDOL [15] fuses frequency domain features with CNNs features via an SE mechanism. By capturing the global response of objects, it enhances the detection performance for objects with large aspect ratios and ultimately achieves 77.22% mAP, while incurring only a limited computational overhead (FPS decreases by 2.9, parameters increase by 0.62 M, and GFLOPs rise modestly). Notably, our proposed SHIFT models instance descriptors from multiple geometric views in a direct manner, achieving the best mAP of 77.95%. While this comes with an increase in GFLOPs (from 134.08 to 197.18), we consider such a tradeoff worthwhile given the significant and consistent gains in detecting challenging elongated objects. In future work, we plan to explore more lightweight designs to further reduce computational costs while preserving performance.

3) Arrangement of Our AFF Module: The standard ResBlock in the ResNet backbone [35] has three convolutional layers. Most recent researches modify the second [37] or third convolutions [38] of ResBlock to boost the performance of the downstream tasks. Here, we name the aforementioned two trends as B_2 and B_3, and separately integrate our AFF into these two convolutional layers. As listed in Table IV, it can be clearly seen that the best performance is always achieved by integrating our AFF module with the second convolutional

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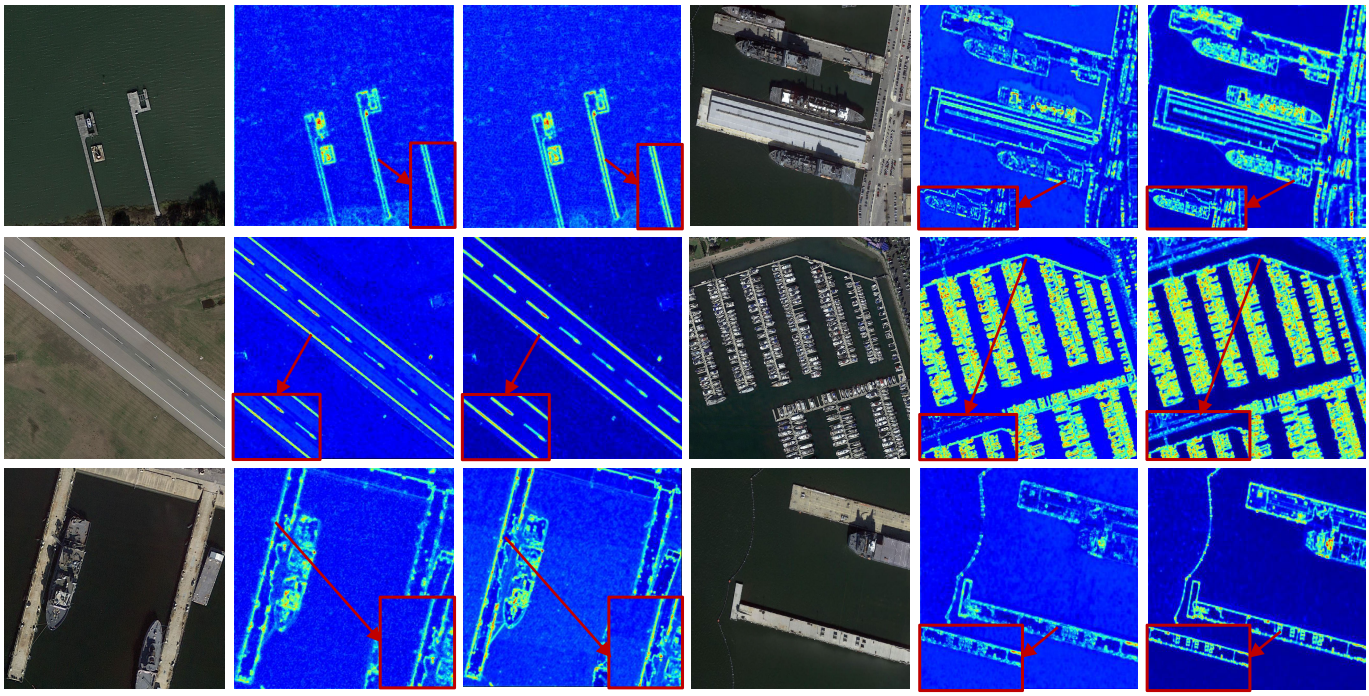


Fig. 6. Visualization of different image features. For each sample image from the DOTA dataset, the original and the SHIFT-enhanced feature are displayed.

TABLE II

COMPARISONS AP50 OF DIFFERENT ASPECT RATIO RELATED METHODS ON DOTA-V1.0. DENOTES THE CORRESPONDING APPROACHES ARE CONSTRUCTED BY OURSELVES BASED ON REDET [7]

Group	Dynamic Sampling		Irregular Kernel		Large Kernel		IoU & Angle Loss		Self-Attention		SHIFT (Ours)
Method | AP50	DRN [21]	72.95	(1×5)	76.92	(5×5)	76.47	R^3Det [29]	73.40	ARS-DETR [25]	75.28	**77.95**
	DCN [27]	76.79	ARC^{50} [4]	77.35	LSKNet-S [6]	77.49	KLD [34]	75.28	SE [24]	76.39	
	DAL [28]	76.95	ARC^{101} [4]	77.70	–	–	B2V [33]	76.21	FDOL [15]	77.22	

TABLE III

COMPARISONS PARAMETERS, FPS, GFLOPS, AND MAP OF DIFFERENT ASPECT RATIO RELATED METHODS ON DOTA-V1.0. ALL THE METHODS ARE BUILT BASED ON REDET [7]. THE FPS OF ALL THE METHODS ARE EVALUATED ON TWO BLOCKS OF GEFORCE GTX 1080TI GPUS

Method	Parameters	FPS	GFLOPs	mAP
ReDet [7]	31.65	7.0	134.08	76.25
+ SE [24]	32.18	5.1	135.01	76.39
+ (5×5)	36.31	3.9	157.33	76.47
+ (1×5)	32.87	4.9	139.56	76.92
+ FDOL [15]	32.27	4.1	142.30	77.22
+ SHIFT (Our)	34.83	4.3	197.18	**77.95**

TABLE IV

PERFORMANCE COMPARISONS OF DIFFERENT ARRANGEMENTS OF OUR AFF ON DOTA-V1.0. CONSIDERING THAT EACH RESBLOCK CONTAINS THREE CONVOLUTIONAL LAYERS, B_i DENOTES THE PROPOSED AFF IS INTEGRATED INTO THE *ith* CONVOLUTIONAL LAYER OF THE RESBLOCK

Method	ResBlock	AP50	AP_{BR}	AP_{HA}
OPLD [46]	–	74.18	53.02	73.23
	B_2	**75.93**	**56.86**	**74.84**
	B_3	75.05	54.70	73.98
Oriented R-CNN [12]	–	76.28	55.27	74.36
	B_2	**77.53**	**57.46**	**76.05**
	B_3	77.08	55.94	75.21
ReDet [7]	–	76.25	53.97	75.96
	B_2	**77.95**	**58.14**	**77.84**
	B_3	76.82	56.06	77.23
ARC [4]	–	77.35	55.33	74.29
	B_2	**78.49**	**57.96**	**78.21**
	B_3	78.12	57.35	77.68

layer of ResBlock, which obtains 1.75%, 1.25%, 1.70%, and 1.14% performance gains for OPLD [46], Oriented R-CNN [12], ReDet [7], and ARC [4], respectively. Importantly, when evaluated within the mAP gain of 1.70% achieved by our method over the ReDet baseline, the AP improvements of 4.17% for BR and 1.88% for HA—both of which are above the average—suggest a relative positive impact on elongated objects.

4) Variants of AFF: We evaluate the performance of different combinations of $\mathcal{A}^f$, $\mathcal{A}^l$, and $\mathcal{A}^t$ on DOTA. As illustrated in Table V, the original SE boosts the baseline method by 0.14% mAP. Importantly, our SHIFT with different variants of AFF always achieves superior performance than the SE algorithm, which attains 0.53%, 0.50%, and 1.56% mAP gains, respectively. The best performance is achieved by merging all the hidden insights into forward-view features, which is 1.70% mAP higher than the baseline method.

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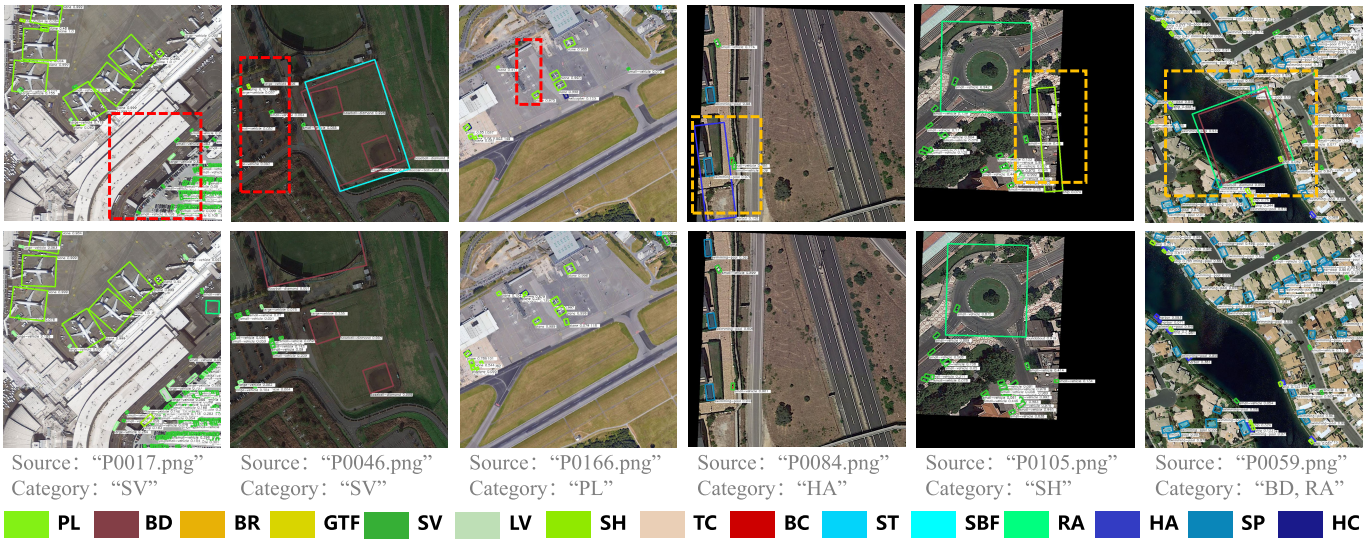


Fig. 7. Qualitative detection results of common aspect ratios objects on DOTA dataset [39]. The first and second rows denote the detection results of the baseline method and our SHIFT. We use red and orange dotted rectangle boxes to highlight the problems of missing and incorrect detection in the baseline method.

TABLE V

PERFORMANCE OF DIFFERENT COMBINATIONS OF $\mathcal{A}^c$, $\mathcal{A}^w$, AND $\mathcal{A}^h$ ON DOTA-V1.0. THE BASELINE METHOD IS REDET [15] IN OUR EXPERIMENTS

Method	$\mathcal{A}^f$	$\mathcal{A}^l$	$\mathcal{A}^t$	AP50
Baseline				76.25
SHIFT (Ours)	✓			76.39
	✓	✓		76.92
	✓		✓	76.89
	✓	✓	✓	**77.95 (+1.70)**

TABLE VI

EFFECTIVENESS OF OUR AFF ON DOTA-V1.0. INDIVIDUAL DENOTES THE EXTRACTED $\mathcal{Y}^l$ OR $\mathcal{Y}^t$ IS REPLACED TO THE ORIGINAL $\mathcal{Y}^f$. MULTIPLE DENOTES FUSING THESE FEATURES FOR AERIAL OBJECT DETECTION

Adaptive combination features ($\mathcal{Y}^f$, $\mathcal{Y}^l$, $\mathcal{Y}^t$)	Fusion Form	AP50
Individual	(1, 0, 0)	76.25
	(0, 1, 0)	68.43
	(0, 0, 1)	69.96
Multiple	*add*	75.38 **(-0.87)**
	concat	75.49 **(-0.76)**
	AFF	**77.95 (+1.70)**

In addition, the mainstream feature aggregation methods use *add* and *concat* operations to boost the representation of image features. In this section, we first explain why the aforementioned manners are not applicable to our approach, and then emphasize the effectiveness of our proposed AFF module. Initially, we directly use $\mathcal{Y}^l$ or $\mathcal{Y}^t$ to replace $\mathcal{Y}^f$ in ResBlock to explore the performance changes. This is what *Individual* represents in Table VI, where 1 in (1, 0, 0) denotes the selected features of ($\mathcal{Y}^f$, $\mathcal{Y}^l$, and $\mathcal{Y}^t$). We can clearly see that feature replacement leads to a rapid decline in detection accuracy. Therefore, we believe that the forms of information represented by features ($\mathcal{Y}^l$ and $\mathcal{Y}^t$) and $\mathcal{Y}^f$ are different. Using *add* and *concat* operations to aggregate multiaxis features is obviously suboptimal. On the contrary, our proposed AFF module achieves 77.95% AP, which attains 1.70%, 1.57%, and 1.46% AP gains over the baseline, *add*-based and *concat*-based methods, respectively. The aforementioned comparisons demonstrate that our AFF module not only bridges the representational gaps among multiaxis features but also models the implicit relationships among these descriptors through the soft attention weights.

D. Comparisons With the State-of-the-Art

1) Results on DOTA-v1.0: As shown in Table VII, we compare our proposed SHIFT with 28 state-of-the-art methods on the DOTA-v1.0 dataset. Under the single-scale testing protocol, our method achieves a competitive mAP of 77.95%. Notably, it delivers significant improvements on two representative large aspect ratio categories, "BR" and "HA," with gains of 4.17% and 1.88% AP over our baseline ReDet [7], respectively. This consistent enhancement of elongated objects validates the core premise of our design. The multiview feature scanning mechanism, enabled by permutation operations, effectively captures the structural cues of objects with extreme geometry that are often under-represented in standard convolutional feature maps.

Furthermore, we evaluate SHIFT under the more challenging multiscale testing setting. Our method attains the best overall performance of 80.26% mAP, surpassing recent strong multiscale detectors such as ACM [40] (79.45% mAP) and MSP-O [61] (79.54% mAP). More importantly, SHIFT establishes new state-of-the-art results on several challenging categories, achieving the highest AP of 61.23% and 72.38% on the large aspect ratio "BR" and the complex "SBF" categories, respectively. These results demonstrate that the proposed multiperspective feature modeling not only excels in a standard setting but also generalizes robustly across scales, confirming its effectiveness in learning more discriminative and geometry-aware representations for aerial object detection.

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TABLE VII

QUANTITATIVE RESULTS OF SINGLE-SCALE AND MULTISCALE OBJECT DETECTION METHODS ON DOTA-V1.0 DATASET. RESULTS OF METHODS ARE OBTAINED FROM THEIR RELEASED CODES. THE BEST AND SECOND-BEST PERFORMANCE OF EACH CATEGORY ARE REPRESENTED BY BOLD AND *

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Single scale																
RoI Transformer [13]	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
DRN [21]	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
SCRDet [47]	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
R^3Det [29]	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
S^2A-Net [48]	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
RQFormer [49]	87.45	78.57	47.36	69.01	79.58	81.27	88.53	90.89	82.80	86.21	58.68	64.20	75.21	74.44	61.39	75.04
AOPG [42]	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.60	66.30	74.05	67.76	58.77	75.39
DODet [50]	89.61	83.10	51.43	72.02	79.16	81.99	87.71	90.89	86.53	84.56	62.21	65.38	71.98	70.79	61.93	75.89
ReDet [7]	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
IBBR [51]	89.55	83.66	54.06	73.93	78.93	83.08	88.29	90.89	86.60	84.80	62.03	65.55	74.16	70.09	58.16	76.25
Oriented R-CNN [12]	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
SANet [52]	89.04	84.66	54.91	78.10	73.58	77.29	87.60	90.88	86.18	84.38	68.45	63.28	76.45	71.40	59.84	76.40
OWSR [16]	**90.41**	85.21	55.00	78.27	76.19	72.19	82.14	90.70	87.22	86.87	66.62	68.43	75.43	72.70	57.99	76.36
DGRL [53]	89.35	84.90	52.58	72.32	78.10	81.90	88.15	90.90	**88.48**	84.52	63.78	66.17	76.44	68.58	60.95	76.48
TSA [54]	89.44	82.95	53.75	73.31	79.29	83.49	87.61	90.90	86.79	84.60	57.48	67.71	73.05	70.65	67.60	76.56
OAN [55]	89.70	84.03	54.61	73.46	79.30	83.27	88.12	90.90	85.22	84.68	62.06	66.87	75.28	70.71	62.75	76.73
B2V [33]	89.31	84.47	53.38	72.67	78.01	82.53	88.07	90.87	87.34	86.14	65.59	65.59	76.24	70.43	65.32	77.06
ARC [4]	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
LSKNet-S [6]	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82	77.49
DynaPro [56]	89.82	80.77	56.90	73.71	80.14	**85.74**	88.44	90.85	86.63	85.60	62.89	70.82	76.38	67.80	71.17	77.85
SHIFT (Ours)	88.82	84.54	58.14	73.98	76.45	84.03	88.12	90.88	87.47	85.50	61.91	63.31	77.84	76.88	71.40	77.95
Multi scale																
DCL [57]	89.26	83.60	53.54	72.76	79.04	82.56	87.31	90.67	86.59	86.98	67.49	66.88	73.29	70.56	69.99	77.37
PSCD [41]	89.86	86.02	54.94	62.02	81.90	85.48	88.39	90.73	86.90	88.82	63.94	69.19	76.84	**82.75**	63.24	78.07
KLD [34]	88.91	85.23	53.64	81.23	78.20	76.99	84.58	89.50	86.84	86.38	71.69	68.06	75.95	72.23	75.42	78.32
OrientedFormer [58]	87.92	83.29	58.92	80.90	81.93	84.62	88.81	90.89	86.81	86.95	66.68	59.86	**78.82**	77.88	71.68	79.06
AMMBA [59]	89.14	82.53	57.80	77.10	**82.67**	85.72	88.53	90.77	87.20	87.82	64.74	69.39	77.72	77.56	69.96	79.24
DDDet [60]	89.55	85.88	57.93	80.38	79.56	83.16	**89.13**	90.85	86.94	87.40	68.51	**71.30**	78.41	79.29	62.84	79.41
ACM [40]	85.55	80.53	61.21	75.40	80.35	85.60	88.32	89.88	87.13	87.10	68.15	67.94	78.75	79.82	75.96	79.45
MSP-O [61]	86.71	**89.18**	58.44	**83.21**	79.26	83.53	87.71	87.13	79.73	**89.52**	61.22	70.31	78.68	81.57	**76.89**	79.54
SHIFT (Ours)	88.86	84.47	**61.23**	80.45	76.99	85.28	88.06	**91.92**	87.37	86.78	**72.38**	66.59	77.80	80.52	75.26	**80.26**

TABLE VIII

QUANTITATIVE RESULTS OF SINGLE-SCALE OBJECT DETECTION METHODS ON DOTA-V1.5 DATASET. RESULTS OF METHODS ARE OBTAINED FROM THEIR RELEASED CODES. THE BEST AND SECOND-BEST PERFORMANCE OF EACH CATEGORY ARE REPRESENTED BY BOLD AND –. * DENOTES THE COPY-PASTE DATA AUGMENTATION [62] IS INTRODUCED

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP
Single scale																	
RTN-O [63]	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR-O [39]	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [64]	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC [65]	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
ReDet [7]	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86
SANet [52]	72.29	76.87	53.25	77.58	52.19	76.34	81.00	90.85	79.36	69.37	68.01	74.07	74.32	66.71	56.44	13.08	67.61
LSKNet-S [6]	72.05	84.94	55.41	74.93	52.42	77.45	81.17	90.85	79.44	69.00	62.10	73.72	77.49	75.29	55.81	42.19	70.26
SHIFT (Ours)	79.08	82.53	56.28	71.28	51.96	75.90	81.05	90.81	76.77	70.39	50.26	72.25	75.06	71.83	70.33	14.23	68.19
SHIFT (Ours)*	79.13	82.25	57.74	73.83	50.26	79.60	82.47	**90.88**	76.26	69.41	56.58	73.56	74.62	73.16	69.57	39.23	70.53
Multi scale																	
GCL [66]	85.88	85.40	45.49	79.57	58.53	73.14	86.52	90.87	81.91	79.03	67.55	76.16	64.70	75.12	71.58	2.42	70.24
ReDet [7]	88.51	**86.45**	61.23	**81.20**	67.60	83.65	**90.00**	90.86	84.30	75.33	71.49	72.06	78.32	74.73	76.10	46.98	76.80
S^2Net + GCL [66]	87.28	85.00	59.42	80.88	**75.90**	83.31	89.85	90.78	**84.90**	**85.67**	69.59	**76.64**	82.08	**78.14**	**79.89**	**49.51**	**78.68**
SHIFT (Ours)	**88.96**	86.05	**62.17**	81.03	67.49	**83.87**	89.54	**90.88**	84.25	78.64	**72.56**	76.01	**83.49**	78.00	77.54	48.12	78.04

2) Analysis of Limitations: In Table VII, we can observe that the AP of "SV" and "LV" of our SHIFT is inferior to many approaches. Considering that these objects exhibit the typical dense distribution characteristics, where our top-view and left-view features tend to be redundant. In contrast, IoU and angle loss-based approaches are more targeted in such scenarios due to their inherent adaptability to geometric constraints and overlap measurement of dense objects. Moreover, the top-view and left-view features cannot fully replace the core role of forward-view features in object representation. This characteristic has a more prominent impact on extremely small-scale objects, e.g., "SV." In our subsequent work, we will further combine multiscale learning and geometric constraints to alleviate the feature representation issues of dense small targets.

3) Results on DOTA-v1.5: Quantitative results of different object detection methods on DOTA-v1.5 are illustrated in Table VIII. First, we find that our SHIFT achieves lower detection performance on "CC" category in DOTA-v1.5,

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TABLE IX

QUANTITATIVE RESULTS OF DIFFERENT OBJECT DETECTION METHODS ON DIOR-R DATASET. RESULTS OF METHODS ARE OBTAINED FROM THEIR RELEASED CODES. THE BEST PERFORMANCE OF EACH CATEGORY IS REPRESENTED BY BOLD

Method	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
RTN-O [63]	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
FR-O [39]	62.79	26.80	71.22	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
Gliding Vertex [67]	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
RoI Transformer [13]	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	78.23	70.26	81.61	54.86	43.27	65.52	63.87
IBBR [51]	63.22	41.39	71.97	88.55	41.23	72.63	28.82	78.90	69.00	70.07	83.01	47.83	55.54	81.23	72.15	62.66	89.05	58.09	43.38	65.36	64.20
AOPG [42]	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99	64.41
DODet [50]	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	71.56	81.52	55.47	51.86	66.40	65.10
GCL [66]	63.08	48.87	72.03	81.33	45.38	80.01	34.16	70.38	80.42	78.41	83.12	47.13	59.42	81.23	73.40	62.47	81.45	56.77	43.85	66.16	65.45
DGRL [53]	69.37	40.99	77.65	89.42	43.97	72.65	30.25	**79.25**	67.28	76.53	**83.13**	45.12	58.01	81.29	**79.55**	62.56	**89.80**	55.71	47.13	64.96	65.73
ReDet [7]	71.03	51.88	71.81	87.92	46.07	78.39	34.90	70.98	77.40	70.88	81.99	**52.54**	57.34	80.32	68.28	69.09	88.24	57.90	47.21	65.19	66.47
DDQ-O [68]	66.71	54.14	73.11	81.80	45.25	77.60	33.37	70.09	79.18	72.24	75.42	47.18	58.27	**89.50**	70.98	74.62	82.83	55.03	56.23	66.61	66.51
DCFL [43]	68.60	53.10	76.70	87.10	42.10	78.60	34.50	71.50	80.80	**79.70**	79.50	47.30	57.40	85.20	64.60	66.40	81.50	58.90	50.90	**70.90**	66.80
TSA [54]	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	66.91
OrientedFormer [58]	65.65	48.69	**78.79**	87.17	41.90	76.34	34.37	72.14	81.40	75.37	79.83	45.15	56.12	88.66	67.59	72.68	87.32	60.31	**56.54**	69.56	67.28
RQFormer [49]	67.31	**55.23**	74.19	82.74	44.49	78.56	**39.85**	70.27	79.84	75.10	80.38	45.64	58.51	88.91	68.10	**75.73**	85.52	57.17	53.54	65.05	67.31
SHIFT (Ours)	**71.23**	49.27	73.06	**89.86**	**48.81**	**80.04**	32.63	75.01	**82.46**	69.85	83.04	51.88	**61.73**	81.31	71.46	70.53	89.51	**60.72**	49.19	65.54	**67.86**

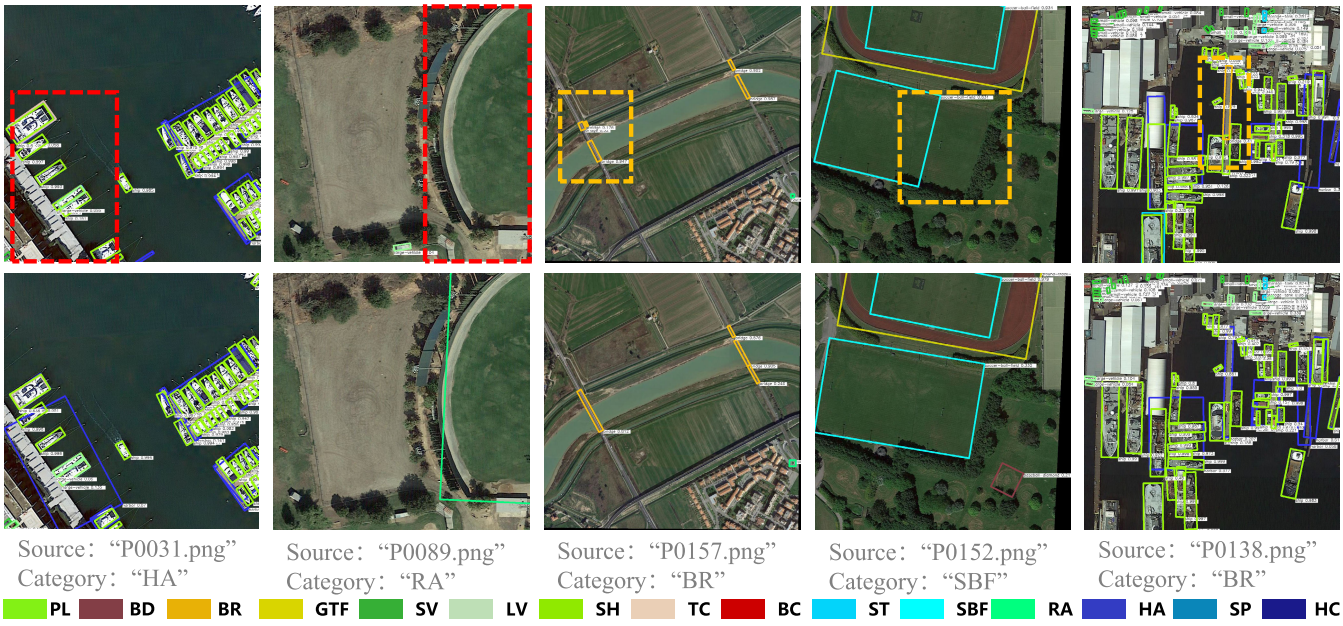


Fig. 8. Qualitative detection results of large aspect ratios objects on DOTA dataset [39]. The first and second rows denote the detection results of the baseline method and our SHIFT. We use red and orange dotted rectangle boxes to highlight the problems of missing and incorrect detection in the baseline method.

which makes our method inferior to the single-scale approach LSKNet-S [6] (14.23% versus 42.19). Considering that the number of CC is much smaller than other classes, we further introduce extra copy-paste data augmentation [62] to boost our SHIFT framework. As listed in Table VIII, the copy-paste data augmentation achieves over 20% AP gains on the CC category, and our SHIFT attains 70.53% mAP. In the future, we will make improvements to our SHIFT in view of the aforementioned limitations to adapt it to the object detection tasks under the condition of sample imbalance.

Compared with the baseline ReDet [7], our method achieves 57.74%, 56.58%, and 74.62% AP on three large aspect ratio categories "BR," "SBF," and "HA," which attains 5.82%, 7.29%, and 1.26% AP gains, respectively. After using the multiscale training and testing strategy, our SHIFT achieves the second-best 78.04% mAP, which attains 7.51% performance gains over the single-scale version of our SHIFT.

4) Results on DIOR-R: We also compare our SHIFT with many detection approaches on the DIOR-R dataset. As listed in Table IX, the proposed SHIFT method achieves the state-of-the-art 67.86% mAP in the whole dataset, and obtains 0.55% improvements over the best competitor RQFormer [49]. Importantly, our method achieves the best AP on 7/20 categories, including many large aspect ratio objects, e.g., BR and BC.

5) Qualitative Results: We first visualize many qualitative results of instances with common aspect ratios on the DOTA dataset [39] in Fig. 7. Obviously, the baseline method not only misses many objects (miss detection), but also prefers generating incorrect bounding boxes to cover instance regions or background regions (incorrect detection), e.g., "HA" and "SH." Importantly, our SHIFT shows its superiority by locating more common aspect ratio objects and generating more accurate bounding boxes to cover the instance regions.

Figs. 8 and 9 illustrate many detection results of instances with large aspect ratios on DOTA and DIOR-R datasets. It can be clearly seen that the baseline method not only misses many large aspect objects, but also predicts incorrect bounding

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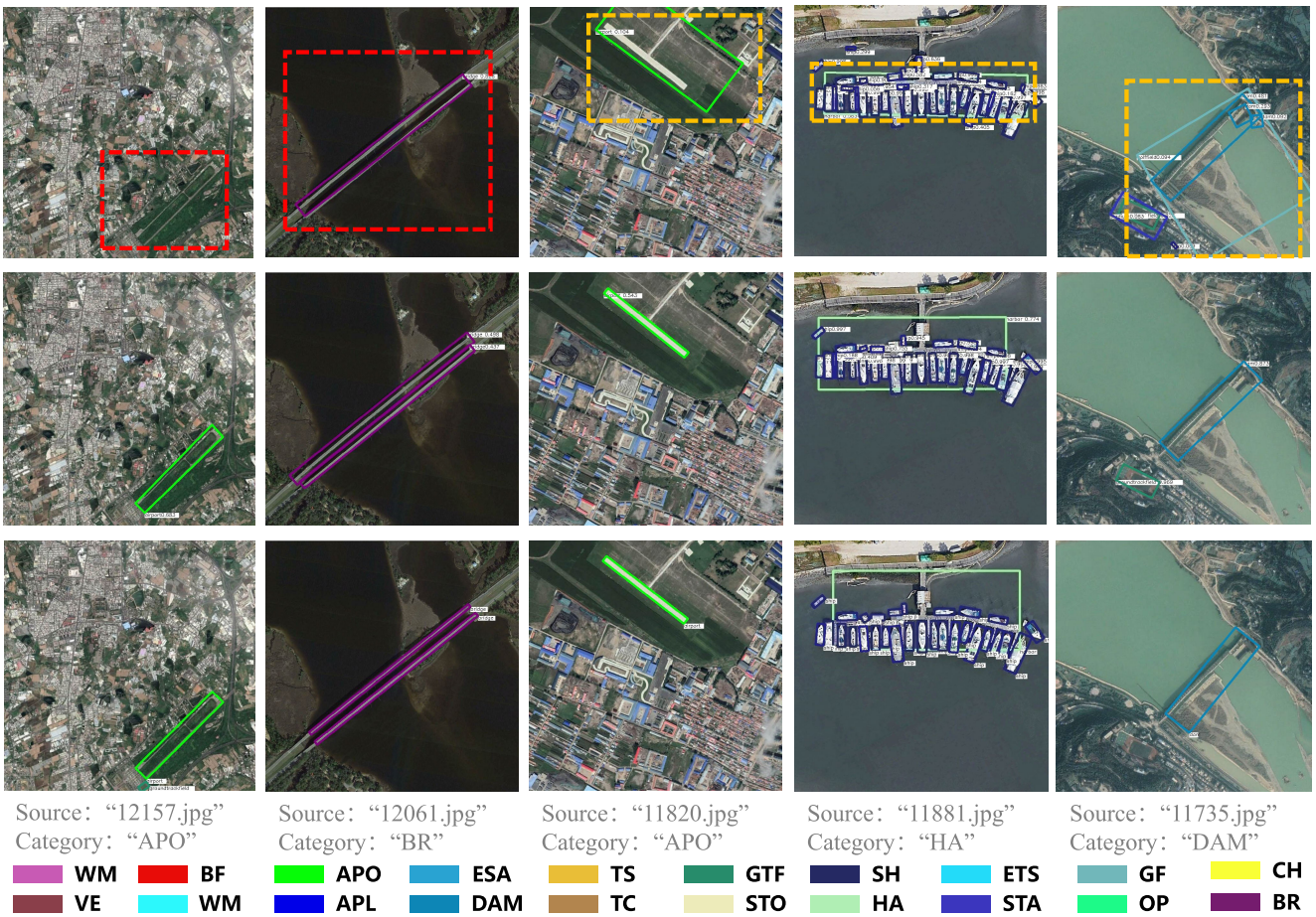


Fig. 9. Qualitative detection results of large aspect ratios objects on DIOR-R dataset [42]. The first and second rows denote the detection results of the baseline method and our SHIFT. We use red and orange dotted rectangle boxes to highlight the problems of missing and incorrect detection in the baseline method.

boxes to cover instance regions, e.g., "HA" and "BR." The aforementioned results prove that the forward-view feature is difficult to perceive instances with large aspect ratios. These comparisons demonstrate that our proposed SHIFT framework is superior to the baseline method on detecting large aspect ratio instances.

V. Conclusion

In this article, we promote the aerial object detection task by paying attention to instances with large aspect ratios. A large aspect ratio object-related method, named SHIFT, is constructed to achieve a more comprehensive understanding of instances in aerial scenarios. First, the common convolution operation is concluded as the process of forward-view feature modeling, which offers comprehensive and professional cues for grasping potential information like the scales and distributions of objects. Moreover, our method further extracts image features from distinct views, e.g., left- and top-views. The aforementioned operations observe objects from different directions, which provide valuable hidden insights for forward-view features. Furthermore, an AFF module is constructed to bridge the gap among these multiaxis features and fuse them in an intelligent way. Extensive experiments prove that our proposed SHIFT network achieves state-of-the-art performance on many challenging benchmarks.

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